CBCS SCHEME

SRI SAIRAM COLLEGE OF ENGINEERING

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BCS503

Fifth Semester B.E./B.Tech. Degree Examination, June/July 2026
Theory of Computation

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	Define DFA. Design the DFA to accept each of the following languages. i) Draw a DFA to accept decimal strings divisible by 3. ii) L = { W∈ {a, b}* , W has even number of a's and odd number of b's}.	10	L2	CO1
	b.	Convert the following NFA to DFA → p {p, q} {p} q {r} {r} r {s} ϕ * s {s} {s}	10	L3	CO1
		OR			
Q.2	a.	Define the following terms with an example : i) Alphabet ii) Power of an alphabet iii) String iv) Concatenation v) Languages.	10	L2	CO1
	b.	Consider the 6 – NFA. Compute ∈ – CLOSURE of each state and convert the automaton to DFA. → p ϕ {p} {q} {r} q {p} {q} {r} ϕ * r {q} {r} ϕ {p}	10	L3	CO1
		Module – 2			
Q.3	a.	Define Regular expression. Obtain the RE for the following languages. i) L = {$a^n b^m$ / m + n is even} ii) L = {w : | w | mod 3 = 0 , where w ∈{a, b}*}.	6	L2	CO2
	b.	Obtain the ∈ - NFA for the Regular expression : i) (a + b)* abb (a + bb) i) ab (a + b)* aab.	6	L2	CO2
	c.	State and prove Pumping Lemma theorem for Regular Languages and prove the Language L = {$a^n b^n$ / n ≥ 1} is not Regular language.	8	L3	CO2
		OR			

Q.4	**a.**	Minimize the following DFA using Table filling algorithm :	**8**	**L3**	**CO2**

	a	b
→A	B	F
B	G	C
* C	A	C
D	C	G
E	H	F
F	C	G
G	G	E
H	G	C

	b.	Prove that Regular Languages are closed under i) Intersection ii) Complementation.	**6**	**L2**	**CO2**
	c.	Obtain a RE for the following (Fig. Q4(c)). Finite Automation by Eliminating states.	**6**	**L2**	**CO2**

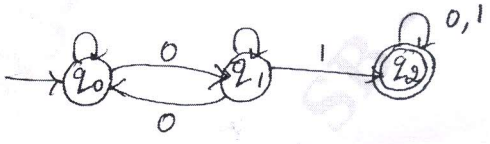


Fig. Q4(c)

Module – 3

Q.5	**a.**	Define Leftmost Derivation , Rightmost Derivation and Parse tree. Construct LMD , RMD and Parse tree for the string aaabbabba using the following grammar. $S \rightarrow a\,B / b\,A$; $A \rightarrow a\,S / b\,A\,A / a$; $B \rightarrow b\,S / a\,B\,B / b.$	**10**	**L2**	**CO3**
	b.	Define Ambiguous grammar. Verify whether the grammar is ambiguous or not for the string id * id + id. $E \rightarrow E + E / E * E / (E) / id$	**10**	**L2**	**CO3**

OR

Q.6	**a.**	Define PDA. Design a PDA to accept the following language: $L = \{a^n b^n \mid n \geq 1\}$.	**10**	**L2**	**CO3**
	b.	Obtain the PDA for the following CFG : $S \rightarrow aABC$ $A \rightarrow aB \mid a$ $B \rightarrow bA \mid b$ $C \rightarrow a$	**10**	**L2**	**CO3**

Module – 4

Q.7	**a.**	Simplify the following CFG by eliminating useless symbols. Remove $\in$ - productions and unit productions : $S \rightarrow AB / C / \in$ $A \rightarrow aA / B / \in$ $B \rightarrow bB / C$ $C \rightarrow c / D$ $D \rightarrow d / \in$ $E \rightarrow e\,E$	**10**	**L2**	**CO4**

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	b.	State and prove Pumping Lemma theorem for CFL's. Show that the language L = { $0^n 1^n 2^n$ / n ≥ 1} is not context free language.	**10**	**L2**	**CO4**
		OR			
Q.8	**a.**	Write the Procedure for Removal of ∈ - productions. Simplify the following grammar : S → aA /aBB A → aAA /∈ B → bB / bbC C → B	**10**	**L3**	**CO4**
	b.	Define CNF. Convert the following grammar to CNF [Chomsky Normal Form]. S → aA /bB A → aAA /bS / b B → bBB / aS / a	**10**	**L3**	**CO4**
		Module – 5			
Q.9	**a.**	Define Turing Machine (TM). Design a TM for the Language L = {$0^n 1^n$ / n > = 1}. Show that the string 0011 is accepted.	**10**	**L3**	**CO5**
	b.	Explain the following : i) Multitape Turing Machine ii) Non- deterministic Turing Machine.	**10**	**L2**	**CO5**
		OR			
Q.10	**a.**	Explain the following : i) Recursively enumerable language. ii) Halting problem in TM.	**10**	**L3**	**CO5**
	b.	Design a TM for the Language L = {$0^n 1^n 2^n$ / n >= 1}. Show that the string 000111222 is accepted.	**10**	**L2**	**CO5**

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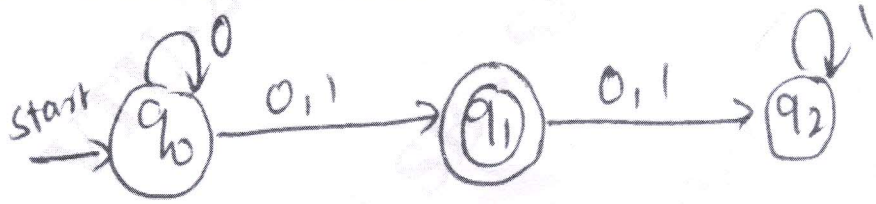
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BCS503

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2025/Jan.2026
Theory of Computation

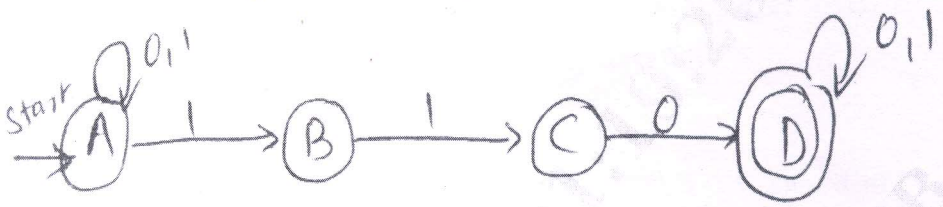
Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	Define the following with example. i) Alphabet ii) String iii) Power of an Alphabet iv) Language v) Problem	05	L2	CO1
	b.	Design DFA for the following languages. i) DFA to accept string of 0's, 1's and 2's beginning with a '0' followed by odd number of 1's and ending with a '2'. ii) L = { W∈ {0,1}* : w does not have 001 as a substring}	08	L3	CO1
	c.	Convert the following NFA to DFA. 	07	L3	CO1
		OR			
Q.2	a.	Design an NFA to recognize the following set of strings 0101, 101 and 011.	05	L2	CO1
	b.	Design an DFA to accept binary numbers divisible by 5.	08	L3	CO1
	c.	Obtain DFA for the following ∈ – NFA δ ∈ a b c →p ϕ {p} {q} {r} q {p} {q} {r} ϕ *r {q} {r} ϕ {p}	07	L3	CO1
		Module – 2			
Q.3	a.	Write the Regular Expression for the following languages. i) $L = \{a^n b^m \mid n >= 4, m <= 3\}$ ii) RE to accept words with two or more letters but beginning and ending with the same letter where ∑ = {a, b}	06	L3	CO2
	b.	Obtain ∈ – NFA for given Regular Expression $(a + b)^* aa (a + b)^*$	06	L3	CO2
	c.	Find the minimized DFA of the following δ a b →A B F B G C *C A C D C G E H F F C G G G E H G C	08	L3	CO2

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OR

Q.4	a.	State and prove pumping theorem for regular languages. Show that the language $L = \{a^n b^n \mid n \geq o\}$ is not regular.	06	L3	CO2
	b.	Convert the following FA to RE using state elimination method. 	06	L3	CO2
	c.	Prove that the Regular Languages are closed under : i) Union ii) Complementation iii) Intersection iv) Difference	08	L3	CO2

Module – 3

Q.5	a.	Design CFG for the following Languages. i) $L = \{ ww^R$ where $W \in \{a, b\}^* \}$ ii) $L = \{ 0^m 1^m 2^n \mid m \geq 1$ and $n \geq 0 \}$ iii) $L = \{ a^n b^{n+3} \mid n \geq 0 \}$	10	L3	CO3
	b.	Consider the grammar G with productions $S \rightarrow aSbS \mid bSaS \mid \in$ Obtain LMD, RMD and parse tree for the string aababb. Is the grammar ambiguous.	10	L3	CO3

OR

Q.6	a.	Obtain PDA to accept the language $L = \{ wCw^R \mid W \in (a + b)^* \}$ and show the moves made by the PDA for the string aabCbaa.	10	L3	CO3
	b.	Convert the following CFG to PDA $S \rightarrow aABC$ $A \rightarrow aB \mid a$ $B \rightarrow bA \mid b$ $C \rightarrow a$	10	L3	CO3

Module – 4

Q.7	a.	Define CNF. Convert the following CFG to CNF $E \rightarrow E + T/T \quad T \rightarrow T * F/F \quad F \rightarrow (E) / I \quad I \rightarrow Ia / Ib \mid a \mid b$	10	L3	CO4
	b.	State and prove pumping lemma for context Free Grammars. Show that $L = \{ 0^n 1^n 2^n \mid n \geq 1 \}$ is not content free.	10	L3	CO4

OR

Q.8	a.	Define CNF convert the following CFG to CNF $S \rightarrow AB \quad A \rightarrow aA \mid bB \mid b \quad B \rightarrow b$	10	L3	CO4
	b.	Prove that the content free languages are closed under i) Union ii) Concatenation iii) Homomorphism	10	L3	CO4

Module – 5

Q.9	a.	Define a Turing Machine. Explain the working of a basic Turing machine with neat diagram.	08	L2	CO5
	b.	Design a Turing Machine to accept the language $L = \{ a^n b^n c^n \mid n \geq 1 \}$. Draw the transition diagram and show the moves made by TM for the string : aabbcc.	12	L3	CO5

OR

Q.10	a.	What are the programming Techniques for Turing Machine. Explain.	10	L2	CO5
	b.	Write short notes on : i) Multi Tape Turing Machine ii) Non Deterministic Turing Machine	10	L2	CO5

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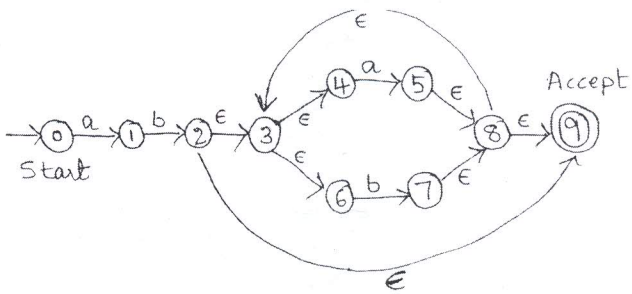
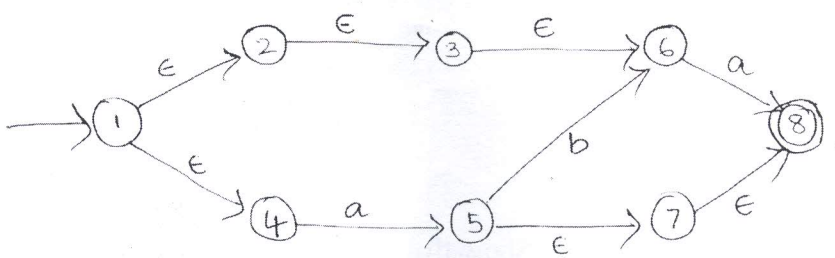
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BCS503

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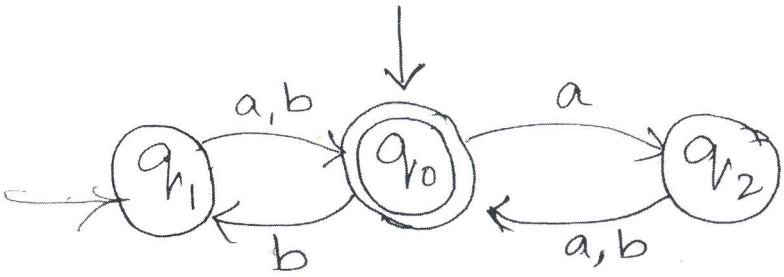
Time: 3 hrs. Max. Marks:100

Note: *1. Answer any FIVE full questions, choosing ONE full question from each module.*
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
1	a.	With a neat diagram, explain a hierarchy of language in automata theory.	5	L2	CO1
	b.	Draw a DFA to accept string of a's and b's ending with ab or ba. Write a transition table for the same.	5	L3	CO1
	c.	Convert the following ϵ– NFA to it's equivalent DFA.  Fig.Q1(c)	10	L3	CO1
		OR			
2	a.	Draw a DFA to accept the languages : i) $L=\{w : n_a(w) \geq 1, n_b(w)=2\}, \Sigma=\{a,b\}$ ii) $L=\{w :\| w \|\bmod 5 \neq 0\}$ on $\Sigma=\{a,b\}$ iii) $L=\{a^p b^q c^r / p \geq o, q > o, r \geq o\}$ on $\Sigma=\{a,b,c\}$.	12	L3	CO1
	b.	Write differences between DFA, NFA and ϵ – NFA.	4	L2	CO1
	c.	Compute ϵ – CLOSURE of each state for the following transition diagram.  Fig.Q2(c)	4	L1	CO1

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Module – 2

3	a.	Write regular expression for the following languages : i) $L=\{a^n b^m / n \geq 0, m \geq o, m+n \text{ is even}\}$ ii) $L=\{a^n b^m / n \geq 1, m > 0, nm \geq 3\}$ iii) $L=\{a^n b^m / n \geq 4, m \leq 3\}$.	9	L3	CO1
	b.	Convert the following automata to a regular expression.  Fig.Q3(b)	3	L2	CO2
	c.	Show that $L=\{a^n b^n / n \geq 0\}$ is not regular.	8	L2	CO2
		OR			
4	a.	Show that regular languages are closed under complementation and intersection.	6	L2	CO2
	b.	Minimize the following DFA and draw the transition diagram of minimized DFA.	8	L3	CO1
	c.	Define Regular Expressions. What are it's applications?	6	L2	CO2

δ	a	b
→A	B	F
B	G	C
* C	A	C
D	C	G
E	H	F
F	C	G
G	G	E
H	G	C

Module – 3

5	a.	Write a grammar for the following languages. i) $L=\{w / n_a(w)=n_b(w)\}$ ii) $L=\{0^i 1^j / i \neq j, i \geq 0, j \geq 0\}$ iii) $L=\{a^{n+2} b^m / n \geq 0, m > n\}$ iv) $L=\{0^m 1^m 2^n / m \geq 1, n \geq 0\}$.	12	L2	CO2
	b.	What is ambiguous grammar? Show that the following grammar is ambiguous. $E \rightarrow E+E / E*E / id$	8	L1	CO2

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		OR			
6	a.	Explain the following terms: i) Pushdown automata (PDA) ii) Languages of PDA iii) Instantaneous description of a PDA.	6	L1	CO2
	b.	Obtain a PDA to accept $L=\{a^n b^n / n \geq 0\}$, Draw the transition diagram for the constructed PDA.	8	L3	CO3
	c.	What is the difference between deterministic and non-deterministic pushdown automata? Explain with an example.	6	L1	CO3
		Module – 4			
7	a.	Consider the grammar $S \rightarrow OA / 1B$ $A \rightarrow OAA / 1S / 1$ $B \rightarrow 1BB / OS / O$ Obtain the grammar in CNF.	10	L3	CO3
	b.	Show that $L=\{a^n b^n c^n / n \geq 0\}$ is not context free.	10	L2	CO3
		OR			
8	a.	State and prove pumping lemma for context free languages.	10	L2	CO3
	b.	Prove that CFL are not closed under complementation and intersection.	10	L2	CO3
		Module – 5			
9	a.	Design a Turing Machine to accept the following languages. $L=\{0^n 1^n 2^n / n \geq 1\}$.	10	L3	CO3
	b.	Write a short notes on i) Multi – tape Turing Machine ii) Non- deterministic TM.	10	L1	CO4
		OR			
10		Write a short notes on i) Post correspondence Problem ii) Halting Problem of TM iii) Recursively enumerable language iv) Linear Bounded Automata.	20	L1	CO4

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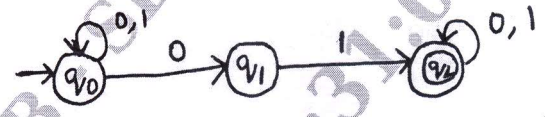
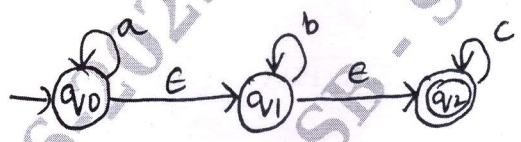
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Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	Define the following with an example: i) Alphabet ii) Power of an alphabet iii) String iv) String concatenation v) Language	5	L1	CO1
	b.	Define Deterministic Finite Automata (DFA) and the language accepted by it.	5	L1	CO1
	c.	Design DFA to accept the following languages: i) $L = \{W \in \{0, 1\}^* : W$ has 001 as a substring$\}$ ii) $L = \{W \in \{0, 1\}^* : \lvert W \rvert \bmod 3 = 0\}$	10	L3	CO1
		OR			
Q.2	a.	Convert the following NFA to DFA  Fig.Q.2(a)	8	L2	CO1
	b.	Convert the following $\in$ - NFA to DFA  Fig.Q.2(b) and define $\in$ - NFA	12	L2	CO1
		Module – 2			
Q.3	a.	Define regular expression. Write the regular expression for the following languages: i) Representing for strings of a and b's having odd length. ii) To accept 10 as substring over an alphabet $\Sigma = \{0, 1\}$	10	L2	CO2
	b.	State and prove pumping Lemma for regular languages.	10	L2	CO2

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		OR			
Q.4	**a.**	Prove that regular languages are closed under complementation and intersection.	**10**	**L2**	**CO2**
	b.	i) Obtain NFA (Non deterministic finite automata) for the regular expression $(a + b)^*$ abb. ii) Obtain NFA for the regular expression $(a^* + ab)(a + b)^*$	**6**	**L2**	**CO2**
	c.	Write the applications of regular expression.	**4**	**L2**	**CO2**
		Module – 3			
Q.5	**a.**	Define context free grammar. Write the CFG for the following languages: i) $L = \left\{ a^n b^n c^m : \begin{matrix} n \geq 0 \\ m >= 0 \end{matrix} \right\}$ ii) $L = \{w \in \{a,b\}^* : n_a(w) = n_b(w)\}$	**10**	**L2**	**CO2**
	b.	i) Define ambigious grammar with suitable example. ii) Consider the grammar $E \rightarrow + EE / * EE / - EE / x / y$ Find the left most derivation, right most derivation and parse tree for the string "+* -xyxy".	**10**	**L2**	**CO2**
		OR			
Q.6	**a.**	Define PDA (Push Down Automata). Design a PDA to accept the following language: $L = \{a^n b^n : n \geq 0\}$. Draw the transition diagram and show that instantaneous description for the string aaabbb.	**10**	**L3**	**CO3**
	b.	Convert the following CFG to PDA: i) $E \rightarrow E + E \mid E * E \mid id$ ii) $E \rightarrow I \mid E * E \mid (E)$ $I \rightarrow id$	**6**	**L2**	**CO3**
	c.	Discuss the language accepted by PDA.	**4**	**L1**	**CO3**
		Module – 4			
Q.7	**a.**	Convert the following grammar to CNF (Chomsky Normal Form) $S \rightarrow ASB / \in$ $A \rightarrow aAS / a$ $B \rightarrow SbS \mid A \mid bb$ and define CNF	**10**	**L2**	**CO3**
	b.	State and prove pumping Lemma for context free languages.	**10**	**L2**	**CO3**

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		OR			
Q.8	**a.**	What are useless and $\in$ productions? Eliminate $\in$, unit and useless productions from the following grammar: $A \rightarrow bA / Bba / aa$ $B \rightarrow aBa / b / D$ $C \rightarrow CA / AC / B$ $D \rightarrow a / \in$	**10**	**L3**	**CO3**
	b.	Prove that the family of context free languages is closed under union concatenation and star closure.	**10**	**L2**	**CO3**
		Module – 5			
Q.9	**a.**	Define a Turing Machine. Explain the working and variants of Turing machine.	**10**	**L1**	**CO4**
	b.	Design a Turing machine to accept $L = \{a^n b^n c^n \mid n \geq 0\}$. Draw the transition diagram. Show the moves mode for string aabbcc.	**10**	**L3**	**CO4**
		OR			
Q.10	**a.**	Explain language acceptability and design of Turing Machines (Steps).	**10**	**L2**	**CO5**
	b.	Explain the following: i) Programming techniques for turing machines ii) Undecidability problem.	**10**	**L2**	**CO5**

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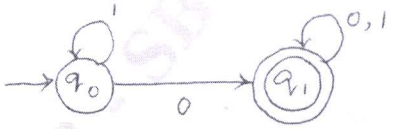
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Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	Define the following with example : i) Language ii) String iii) Power of an alphabet.	3	L1	CO1
	b.	Define DFA. Draw a DFA to accepts. i) The set of all strings that contain a substring aba. ii) To accept the stings of a's and b's that contain not more than there b's. iii) L = {w ∈ {a, b}* : No 2 consecutive characters are same in w}.	10	L3	CO1
	c.	Convert the following NFA to DFA. → p {p, q} {p} q {r} {r} r {s} ϕ * s {s} {s}	7	L2	CO1
		OR			
Q.2	a.	Define the following with example : i) Alphabet ii) Reversal of string iii) Concatenation of Languages.	3	L1	CO1
	b.	Design a DFA for the Language : L = {w ∈ {0, 1}* : w is a string divisible by 5}.	7	L3	CO1
	c.	Define NFA. Obtain an ε - NFA which accepts strings consisting of 0 or more a's , followed by 0 or more b's followed by 0 or more C's. Also convert it to DFA.	10	L2	CO1
		Module – 2			
Q.3	a.	Define Regular expression. Write the regular expression for the following languages : i) Strings of a's and b's starting with a and ending with b. ii) Set of strings that consists of alternating 0's and 1's. iii) L = {a^n bm , (n + m) is even}. iv) L = {w : / w / mod 3 = 0 , where w ∈{a, b}*}.	10	L2	CO2

	b.	Minimize the following finite automata using Table filling algorithm : δ | a | b → A | B | A B | A | C C | D | B * D | D | A E | D | F F | G | E G | F | G H | G | D	10	L2	CO2
		OR			
Q.4	a.	Construct ε - NFA for the following Regular expression : i) $(0+1)\,0\,1(1+0)$ ii) $1(0+1)^*\,0$ iii) $(0+1)^*\,0\,1\,1^*$	6	L1	CO2
	b.	Obtain the Regular expression that denotes the language accepted by Fig. Q4(b). Fig. Q4(b)  Using Kleene's theorem.	6	L3	CO2
	c.	State the Pumping Lemma for the Regular Languages. And also prove that the following languages are note regular. i) $L = \{0^n\,1^m \mid n \le m\}$ ii) $L = \{0^n\,1^m\,2^n \mid n, m \ge 1\}$.	8	L1	CO2
		Module – 3			
Q.5	a.	Design CFG for the following languages : i) $L = \{a^n\,b^{n+3}, n \ge 0\}$ ii) $L = \{a^i\,b^j\,c^k, j = i + k, i \ge 0, k \ge 0\}$ iii) $L = \{w / /w/ \bmod 3 > 0 \text{ where } w \in \{a\}^*\}$ iv) $L = \{a^m\,b^n / m \ne n\}$ v) Palinderomes over 0 and 1.	10	L3	CO3
	b.	Consider the grammar G with productions. $S \rightarrow A\,b\,B / A / B$; $A \rightarrow aA / \varepsilon$; $B \rightarrow a\,B / b\,B / \varepsilon$. Obtain LMD , RMD and parse tree for the string aaabab. Is the given grammar ambiguous?	10	L2	CO3
		OR			
Q.6	a.	Define the following with example : i) Context free grammar ii) Left most Derivation iii) Parse tree iv) Ambiguous grammar.	4	L1	CO3
	b.	Design PDA for the language : $L = \{a^i\,b^j\,c^k / i + k = j, i \ge 0, k \ge 0\}$ and show the moves made by the PDA for the string aabbbc.	10	L3	CO3

	c.	Convert the following CFG's to PDA : $S \rightarrow aA$; $A \rightarrow a\,ABC / bB / a$; $B \rightarrow b$; $C \rightarrow c$.	6	L2	CO3
		Module – 4			
Q.7	a.	Define CNF. Convert the following CFG to CNF $E \rightarrow E + T / T$ $T \rightarrow T * F / F$ $F \rightarrow (E) / I$ $I \rightarrow Ia / Ib / a / b$.	10	L2	CO4
	b.	Show that $L = \{0^n 1^n 2n / n \geq 1\}$ is no context free.	4	L2	CO4
	c.	Prove that the family of context free languages is closed under union and concatenation.	6	L1	CO4
		OR			
Q.8	a.	Define Greibach Normal Form. Convert the following CFG to GNF. $S \rightarrow AB$; $A \rightarrow aA / bB / b$; $B \rightarrow b$.	6	L2	CO4
	b.	Consider the following CFG : $S \rightarrow ABC / BaB$ $A \rightarrow aA / BaC / aaa$ $B \rightarrow bBb / a / D$ $C \rightarrow CA / AC$ $D \rightarrow \varepsilon$ i) What are useless symbols? ii) Eliminate ε - productions , Unit productions and useless symbols from the grammar.	10	L3	CO4
	c.	Prove that the following languages are not context free. i) $L = \{ai / i \text{ is prime}\}$ ii) $L = \{a^{n^2} / n \geq 1\}$.	4	L2	CO3
		Module – 5			
Q.9	a.	Define a turing machine and explain with neat diagram, the working of a basic turing machine.	6	L1	CO4
	b.	Design a Turing machine to accept the language, $L = \{a^n b^n c^n / n \geq 1\}$. Draw the transition diagram and show the moves for the string aabbcc.	14	L4	CO4
		OR			
Q.10	a.	Design a Turing machine to accept palindrome over {a, b} and draw the transition diagram.	12	L4	CO5
	b.	Write a short notes on : i) Recursively Enumerable Language. ii) Multitape Turing Machine.	8	L1	CO5

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